

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET E — CASE STUDY & PRACTICAL EXPERIMENTS

CHAPTER 04: DESCRIBING MOTION & KINEMATICS

TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-E-04

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET E - CASE STUDY & PRACTICAL EXPERIMENTS):

1. The question paper comprises **5 In-Depth Practical Case Studies** with **5 MCQs each (Total: 25 Questions, 1 Mark each)**.
2. Read each laboratory experiment, data table, and passage carefully before answering the corresponding questions.
3. Each question has 4 options with a single correct answer.
4. All questions are compulsory. There is no negative marking.
5. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

CASE STUDY 1: TICKER-TIMER ANALYSIS OF LINEAR ACCELERATION ON AN INCLINE

Students set up an aluminum track inclined at an angle to the bench. A dynamic trolley was released from rest at the top of the ramp, pulling a paper tape through an electromagnetic ticker-tape timer vibrating at a frequency of 50 Hz (producing 50 dots per second).

- Time between two consecutive dots = $\frac{1}{50} \text{ s} = 0.02 \text{ s}$.

The students analyzed a segment of tape where dot positions were clearly marked:

- Distance between Dot 0 and Dot 5 ($t_1 = 0.10 \text{ s}$) = 2.0 cm
- Distance between Dot 5 and Dot 10 ($t_2 = 0.10 \text{ s}$) = 6.0 cm
- Distance between Dot 10 and Dot 15 ($t_3 = 0.10 \text{ s}$) = 10.0 cm
- Distance between Dot 15 and Dot 20 ($t_4 = 0.10 \text{ s}$) = 14.0 cm

Q.1 What is the average velocity of the trolley during the time interval between Dot 0 and Dot 5 (0.00 s to 0.10 s)?

- (A) 0.10 m/s (B) 0.20 m/s
(C) 0.40 m/s (D) 0.02 m/s

Q.2 What is the average velocity of the trolley during the interval between Dot 5 and Dot 10 (0.10 s to 0.20 s)?

- (A) 0.20 m/s (B) 0.60 m/s
(C) 0.40 m/s (D) 0.80 m/s

Q.3 Calculate the constant acceleration (a) of the trolley down the inclined track:

- (A) 2.0 m/s^2 (B) 4.0 m/s^2
(C) 0.4 m/s^2 (D) 9.8 m/s^2

Q.4 What indicates that the trolley is moving with uniform acceleration rather than constant velocity?

- (A) The dots on the tape are spaced at equal distances
(B) The distance between successive sets of dots increases by a constant difference of 4.0 cm every 0.10 s
(C) The tape moves backward
(D) The timer frequency fluctuates

Q.5 What would the velocity-time ($v-t$) graph for this trolley look like?

- (A) A horizontal line parallel to the time axis
(B) A straight line sloping upwards with a positive slope of 4.0 m/s^2
(C) A curve with decreasing slope
(D) A vertical line perpendicular to the time axis

CASE STUDY 2: AUTOMOTIVE BRAKING DISTANCE & DRIVER REACTION DYNAMICS

A traffic safety research institute analyzed the total stopping distance of a car travelling at different initial speeds (u) on a dry asphalt highway.

- Total Stopping Distance consists of two phases:

1. **Thinking Distance (d_{think}):** Distance covered during the driver's physiological reaction time ($t_r = 0.60 \text{ s}$) before the brakes are engaged ($d_{\text{think}} = u \times t_r$).

2. **Braking Distance (d_{brake}):** Distance covered while the brakes apply a constant retarding deceleration of $a = -5.0 \text{ m/s}^2$ until the car comes to rest ($v=0$).

- Data for two test speeds:
- Car A speed: $36 \text{ km/h} = 10 \text{ m/s}$
- Car B speed: $72 \text{ km/h} = 20 \text{ m/s}$

Q.6 What is the thinking distance covered by Car A ($u = 10 \text{ m/s}$) before the driver presses the brake pedal?

- (A) 3.0 m
(B) 6.0 m
(C) 10.0 m
(D) 12.0 m

Q.7 What is the actual braking distance (d_{brake}) required for Car A to come to a complete stop?

- (A) 5.0 m
(B) 10.0 m
(C) 20.0 m
(D) 15.0 m

Q.8 What is the total stopping distance (Thinking + Braking) for Car A?

- (A) 10.0 m
(B) 16.0 m
(C) 20.0 m
(D) 26.0 m

Q.9 For Car B travelling at double the speed ($72 \text{ km/h} = 20 \text{ m/s}$), what is its braking distance (d_{brake})?

- (A) 20.0 m
(B) 40.0 m
(C) 30.0 m
(D) 80.0 m

Q.10 Why does braking distance depend quadratically on initial speed ($d_{\text{brake}} \propto u^2$)?

- (A) Because car mass doubles with speed
(B) Because initial kinetic energy is proportional to u^2 , and the braking work ($F \times d$) must dissipate this kinetic energy
(C) Because reaction time increases with speed
(D) Because gravity increases with speed

CASE STUDY 3: FREE-FALL KINEMATICS & PRECISION PHOTOGATE MEASUREMENT OF g

In a modern school physics lab, students measured the acceleration due to gravity (g) using an automated free-fall apparatus equipped with two infrared photogates connected to a millisecond digital timer.

- A steel sphere of mass 28 g was held by an electromagnet at height h directly above Photogate 1 ($s=0$, $t=0$).
- When released from rest ($u=0$), the ball interrupted Photogate 1 (starting the timer) and interrupted Photogate 2 positioned at a vertical distance h below.
- **Trial 1:** Vertical fall height $h = 0.450 \text{ m}$ $\rightarrow$ Measured fall time $t = 0.303 \text{ s}$
- **Trial 2:** Vertical fall height $h = 0.800 \text{ m}$ $\rightarrow$ Measured fall time $t = 0.404 \text{ s}$
- **Trial 3:** Vertical fall height $h = 1.250 \text{ m}$ $\rightarrow$ Measured fall time $t = 0.505 \text{ s}$

Q.11 Which kinematic equation directly relates the vertical fall height h and time t for a body dropped from rest?

- (A) $h = ut + \frac{1}{2}gt^2$ which simplifies to $h = \frac{1}{2}gt^2$ (B) $v = u + gt$
 (C) $v^2 = 2gh$ (D) $h = \frac{u+v}{t}$

Q.12 Using the data from Trial 2 ($h = 0.800 \text{ m}$, $t = 0.404 \text{ s}$), calculate the experimental value of g :

- (A) 9.80 m/s^2 (B) 9.65 m/s^2
 (C) 10.2 m/s^2 (D) 9.40 m/s^2

Q.13 What is the instantaneous velocity of the steel sphere as it crosses Photogate 2 in Trial 3 ($h = 1.250 \text{ m}$, take $g = 9.80 \text{ m/s}^2$)?

- (A) 4.95 m/s (B) 2.45 m/s
 (C) 7.00 m/s (D) 9.80 m/s

Q.14 If the 28 g steel sphere is replaced by a massive 150 g steel sphere dropped from the exact same height, how does the fall time change?

- (A) Fall time decreases to one-fifth (B) Fall time increases by five times
 (C) Fall time remains virtually identical (D) The heavier sphere stops mid-air

Q.15 What would a graph of h versus t^2 yield for a freely falling object?

- (A) A downward parabola (B) A straight line passing through the origin with slope equal to $\frac{1}{2}g$
 (C) A hyperbolic curve (D) A horizontal line

CASE STUDY 4: MULTI-STAGE VELOCITY-TIME GRAPH ANALYSIS OF AN EXPRESS TRAIN

An express train travels along a straight railway track between two stations, Station P and Station Q. The graph of its velocity (v in m/s) against time (t in seconds) is recorded over a 60-second journey:

- **Stage 1** ($0 \text{ to } 20 \text{ s}$): Velocity increases uniformly from rest (0 m/s) to 30 m/s .
- **Stage 2** ($20 \text{ to } 50 \text{ s}$): The train maintains a constant cruise velocity of 30 m/s .
- **Stage 3** ($50 \text{ to } 60 \text{ s}$): The brakes are applied uniformly, bringing the train to rest at 60 s ($v=0$).

Q.16 What is the acceleration of the train during Stage 1 ($0 \text{ to } 20 \text{ s}$)?

- (A) 1.5 m/s^2 (B) 3.0 m/s^2
 (C) 0.75 m/s^2 (D) 2.0 m/s^2

Q.17 What is the acceleration of the train during Stage 2 (20 m/s to 50 m/s)?

- (A) 1.5 m/s^2 (B) 0 m/s^2
(C) 30 m/s^2 (D) -3.0 m/s^2

Q.18 What is the retardation (deceleration) of the train during Stage 3 (50 m/s to 60 m/s)?

- (A) -1.5 m/s^2 (B) 3.0 m/s^2
(C) 6.0 m/s^2 (D) 0.3 m/s^2

Q.19 What is the total displacement (distance between Station P and Station Q) covered by the train in 60 seconds?

- (A) 1200 m (B) 1350 m
(C) 1500 m (D) 1800 m

Q.20 What is the average speed of the train over the entire 60-second journey?

- (A) 22.5 m/s (B) 30.0 m/s
(C) 15.0 m/s (D) 25.0 m/s

CASE STUDY 5: CENTRIPETAL MECHANICS IN UNIFORM CIRCULAR MOTION

In a physics laboratory investigation of circular motion, students whirled a rubber bung of mass $m = 0.050\text{ kg}$ tied to a nylon string of radius $r = 0.80\text{ m}$ in a horizontal circular path.

- The string passed through a smooth glass tube, and its lower end supported a hanging counterweight of mass $M = 0.40\text{ kg}$.
- The students maintained steady horizontal rotation such that the hanging mass remained stationary at equilibrium ($T = Mg = 0.40 \times 9.8 = 3.92\text{ N}$).
- The rubber bung completed 20 revolutions in 20.0 seconds (period $T_{\text{rev}} = 1.0\text{ s}$).

Q.21 What is the linear speed (v) of the rubber bung moving along the circular path?

- (A) 2.51 m/s (B) 5.03 m/s
(C) 1.60 m/s (D) 10.0 m/s

Q.22 What is the magnitude of the centripetal acceleration (a_c) experienced by the rotating bung?

- (A) 6.28 m/s^2 (B) 31.6 m/s^2
(C) 15.8 m/s^2 (D) 9.80 m/s^2

Q.23 What provides the necessary centripetal force to keep the rubber bung moving in a circle?

- (A) The tension force in the nylon string (B) Gravitational force of the Earth pulling downward
(C) Air friction pushing the bung forward (D) Magnetic attraction from the glass tube

Q.24 In which direction does the rubber bung fly if the nylon string snaps suddenly?

- (A) Radially inward toward the glass tube (B) Radially outward away from the center
(C) Tangentially along a straight line in the direction of its instantaneous velocity (D) It stops and drops vertically downward

Q.25 Why is uniform circular motion classified as an accelerated motion even when its speed is strictly constant?

- (A) Because its mass is changing (B) Because the direction of the velocity vector is continuously changing at every point along the circular path
- (C) Because the radius increases continuously (D) Because time slows down during rotation

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET E)

01	(A) (B) (C) (D)	02	(A) (B) (C) (D)	03	(A) (B) (C) (D)	04	(A) (B) (C) (D)	05	(A) (B) (C) (D)
06	(A) (B) (C) (D)	07	(A) (B) (C) (D)	08	(A) (B) (C) (D)	09	(A) (B) (C) (D)	10	(A) (B) (C) (D)
11	(A) (B) (C) (D)	12	(A) (B) (C) (D)	13	(A) (B) (C) (D)	14	(A) (B) (C) (D)	15	(A) (B) (C) (D)
16	(A) (B) (C) (D)	17	(A) (B) (C) (D)	18	(A) (B) (C) (D)	19	(A) (B) (C) (D)	20	(A) (B) (C) (D)
21	(A) (B) (C) (D)	22	(A) (B) (C) (D)	23	(A) (B) (C) (D)	24	(A) (B) (C) (D)	25	(A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & PRACTICAL INVESTIGATION EXPLANATIONS (SET E)

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.1	(B)	$v_1 = \frac{\Delta s_1}{\Delta t} = \frac{2.0 \text{ cm}}{0.10 \text{ s}} = 20.0 \text{ cm/s} = 0.20 \text{ m/s}$.
Q.2	(B)	$v_2 = \frac{6.0 \text{ cm}}{0.10 \text{ s}} = 60.0 \text{ cm/s} = 0.60 \text{ m/s}$.
Q.3	(B)	$a = \frac{v_2 - v_1}{\Delta t} = \frac{0.60 - 0.20}{0.10} = \frac{0.40 \text{ m/s}}{0.10 \text{ s}} = 4.0 \text{ m/s}^2$.
Q.4	(B)	Equal increments in displacement during successive equal time intervals ($\Delta s = 4.0 \text{ cm}$ increase per interval) confirm constant acceleration.
Q.5	(B)	For uniform acceleration, velocity increases linearly with time ($v = u + at$), yielding a straight line with gradient equal to acceleration.
Q.6	(B)	$d_{\text{think}} = u \times t_r = 10 \text{ m/s} \times 0.60 \text{ s} = 6.0 \text{ m}$.
Q.7	(B)	Using $v^2 = u^2 + 2as \rightarrow 0^2 = 10^2 + 2(-5.0)d_{\text{brake}} \rightarrow 10 d_{\text{brake}} = 100 \rightarrow d_{\text{brake}} = 10.0 \text{ m}$.
Q.8	(B)	Total Stopping Distance $= d_{\text{think}} + d_{\text{brake}} = 6.0 \text{ m} + 10.0 \text{ m} = 16.0 \text{ m}$.
Q.9	(B)	$d_{\text{brake}} = \frac{u^2}{2a} = \frac{20^2}{2 \times 5.0} = \frac{400}{10} = 40.0 \text{ m}$. Notice that doubling speed quadruples braking distance ($10 \times 4 = 40 \text{ m}$).
Q.10	(B)	According to the work-energy theorem, braking work equals kinetic energy: $F_{\text{brake}} \times d = \frac{1}{2} m u^2 \rightarrow d \propto u^2$.
Q.11	(A)	Since the ball is released from rest ($u = 0$), $s = ut + \frac{1}{2}at^2$ reduces directly to $h = \frac{1}{2}gt^2$.
Q.12	(A)	$g = \frac{2h}{t^2} = \frac{2 \times 0.800}{(0.404)^2} = \frac{1.600}{0.1632} \approx 9.80 \text{ m/s}^2$.
Q.13	(A)	$v = \sqrt{2gh} = \sqrt{2 \times 9.80 \times 1.250} = \sqrt{24.5} \approx 4.95 \text{ m/s}$.
Q.14	(C)	All bodies, regardless of mass, experience the identical gravitational acceleration ($a=g$) when aerodynamic drag is negligible.
Q.15	(B)	Since $h = \left(\frac{1}{2}g\right) t^2$, plotting h on y-axis against t^2 on x-axis produces a straight line with slope $m = \frac{1}{2}g$.
Q.16	(A)	$a_1 = \frac{v - u}{\Delta t} = \frac{30 - 0}{20} = 1.5 \text{ m/s}^2$.
Q.17	(B)	Velocity is constant at 30 m/s; therefore, rate of change of velocity is zero ($a = 0$).
Q.18	(B)	$a_3 = \frac{0 - 30}{10} = -3.0 \text{ m/s}^2$. Retardation is the magnitude of deceleration, which is 3.0 m/s^2 .
Q.19	(B)	Total displacement equals the area of the trapezium beneath the v - t graph: $\text{Area} = \frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height} = \frac{1}{2} \times (60 + 30) \times 30 = \frac{1}{2} \times 90 \times 30 = 1350 \text{ m}$.
Q.20	(A)	$\text{Average speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{1350 \text{ m}}{60 \text{ s}} = 22.5 \text{ m/s}$.
Q.21	(B)	$v = \frac{2\pi r}{T_{\text{rev}}} = \frac{2 \times 3.1416 \times 0.80 \text{ m}}{1.0 \text{ s}} = 5.026 \text{ m/s} \approx 5.03 \text{ m/s}$.
Q.22	(B)	$a_c = \frac{v^2}{r} = \frac{(5.026)^2}{0.80} = \frac{25.27}{0.80} \approx 31.6 \text{ m/s}^2$.
Q.23	(A)	The inward radial tension in the string pulls the revolving bung toward the center of rotation, providing the centripetal force.
Q.24	(C)	The moment centripetal tension disappears, the bung continues along its instantaneous tangential straight-line velocity due to inertia.

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.25	(B)	Velocity is a vector quantity. Even if its magnitude (speed) is constant, continuous turning of its direction produces continuous centripetal acceleration.